

CHAPTER 10

From Grounded Evidence to a Defensible Report

10

What Does a Good Validation Report Have to Get Right?

Three different jobs: include the required work, support the claims and explain the results clearly.

COVERAGE

Did we report every required analysis, adverse result and limitation?

GROUNDING

Can every factual statement and number be traced to validation evidence?

WRITING

Can a reader understand the evidence, comparison and qualification?

Good writing cannot repair missing evidence or an omitted finding.

Build the Evidence Packet Before Writing

For each report section, decide what evidence it needs and retrieve that evidence first.



A PERFORMANCE PACKET MIGHT CONTAIN

Confirmation result • comparator • affected population • interval • limitations • source passages

IF THE QUERY DOES NOT RESOLVE

Record the status: missing evidence, ambiguous binding or conflicting records. Do not let the LLM fill the gap.

Evidence is selected before generation, not by the writing model while it is drafting.

What Does the LLM Receive?

The writing prompt is small and explicit: rules + section requirements + evidence.

COMMON RULES

Use supplied evidence only

Preserve qualifications

Do not invent support

Keep scope and units intact

SECTION REQUIREMENTS

What this section must cover

Required comparisons

What conclusions are allowed

EVIDENCE PACKET

Confirmed findings

Exact values

Source passages

Known gaps and limitations

The LLM writes from the packet. It does not decide what the evidence establishes.

Check the Draft in Both Directions

One check catches unsupported claims. The other catches missing content.

DRAFT → EVIDENCE

Take each claim in the draft and ask:

- Is it supported?
- Is the number exact?
- Is the scope correct?
- Did the draft overstate the result?

This is the faithfulness check.

REQUIREMENTS → DRAFT

Take each required item and ask:

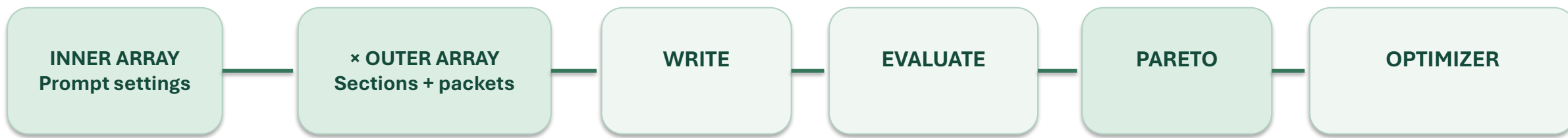
- Did the draft include it?
- Did it include adverse results?
- Did it preserve limitations?
- Did it report unresolved gaps?

This is the completeness check.

A report can contain no false statements and still be wrong because it left something important out.

Prompt Optimization Is Another Discovery Loop

Keep the evidence fixed. Search for a better way to instruct the writer.



INNER ARRAY — WHAT WE VARY

Instruction order • section structure • level of detail • citation rules • qualification rules

EVALUATOR — WHAT WE MEASURE

Faithfulness • completeness • qualification preservation • readability

We are optimizing the writing instruction, not changing the findings.

Reflection Helps When the Prompt Space Is Missing Something

The optimizer searches the prompt factors we already defined. Reflection can propose a new one.

PATTERN IN THE FAILURES

Drafts are faithful, but synthesis sections repeatedly drop important limitations.

REFLECTION

Propose a new instruction:

Every reported result must carry its attached exception or limitation.

TEST THE REVISION

Admit the new prompt rule, rebuild the instruction and evaluate it on the same development cases.

Reflection proposes the change. Measurement decides whether it helped.

Confirm the Selected Prompt on Reserved Sections

Prompt search creates selection bias too. Use the Chapter 9 logic.

SEARCH

Use development sections to compare prompt candidates and select one.

FREEZE

Freeze the selected instruction, evidence-binding rules and writing-model settings.

CONFIRM

Compare the selected prompt with the baseline on reserved sections using the same quality measures.

If reserved results change the prompt, those sections are no longer confirmation evidence.

From Confirmed Findings to the Final Report

Chapter 10 turns the evidence from earlier chapters into a report that can be checked.



WHAT THE REPORT MAY SAY

Only what the evidence supports, with the same population, period, uncertainty and limitations.

WHAT THE REPORT MUST SAY

Everything required by the approved schema, including adverse results and unresolved gaps.

The report should preserve the validation evidence, not improve the story.